bdscrub 1.0.0

Cleaning Bangladesh-specific data fields in R — reference guide with examples

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1. What bdscrub does

Survey and administrative data from Bangladesh have the same problems again and again: mobile numbers typed in many formats, often two per cell; money written as “1.5 lakh”, “Tk 25,000/-” or “15-20k”; Bangla digits; missing values written as “N/A”, “-” or “#N/A”; and place names spelled differently in every source, and differently again in the boundary (shape or GeoJSON) files used for maps, so that joins silently lose rows. bdscrub handles these with **eleven functions**:

Function	What it does	Typical call
`bd_phone()`	mobile numbers in one format; several numbers per cell; operator	`bd_phone(df$mobile)`
`bd_amount()`	money amounts with lakh, crore, Bangla digits; ranges	`bd_amount(df$income, details = TRUE)`
`bd_number()`	plain numbers (age, household size)	`bd_number(df$age)`
`bd_taka()`	write amounts with lakh/crore grouping	`bd_taka(x, prefix = "Tk ")`
`bd_digits()`	Bangla digits to English digits and back	`bd_digits(x)`
`bd_place()`	district, division and thana names (upazila names too); codes; map spellings	`bd_place(df$thana, "thana", district = df$district)`
`bd_address()`	post code, thana, district, division from an address	`bd_address(df$address)`
`bd_missing()`	“N/A”, “-”, “#N/A” … to `NA`	`bd_missing(x)`
`bd_squish()`	extra and invisible spaces	`bd_squish(x)`
`bd_check()`	how many values of each column can be read, and why not	`bd_check(df, spec)`
`bd_clean()`	clean many columns at once	`bd_clean(df, spec)`

The package works offline and needs only base R (R ≥ 4.1). Tables are returned as tibbles when the `tibble` package is installed, otherwise as data frames. The output in this guide was produced without tibble, so tables print as data frames; with tibble the values are the same.

2. What is in the v1_final folder, and how to start

Item	What it is
`bdscrub/`	The package source (R code, documentation, data, tests, vignette). Install with `devtools::install("bdscrub")` or build it in RStudio.
`bdscrub_1.0.0.tar.gz`	The built package (if present): `install.packages("bdscrub_1.0.0.tar.gz", repos = NULL, type = "source")`.
`bdscrub_standalone.R`	The whole package in one R file, for computers where packages cannot be installed: `source("bdscrub_standalone.R")`. Generated from the package by `bdscrub/tools/make_standalone.R`.
`README.pdf`	This guide.

Item	What it is
`data/`	`bdscrub_messy_survey.csv` (3,030 messy survey rows), `bdscrub_answer_key.csv` (the true value of every cell) and `make_messy_data.py` (the script that made both).
`demo/demo_bdscrub.R`	Runs every function on the messy data and scores the results against the answer key.
`slides/bdscrub_v1_slides.html`	Presentation: the functions, how they work, and the demonstration. Open in a browser; arrow keys to move.
`bdscrub_v1_bundle.zip`	All of the above in one file, to share.

> **Quick start.** In RStudio: `devtools::install("path/to/v1_final/bdscrub")`, then open `demo/demo_bdscrub.R`, use *Session > Set Working Directory > To Source File Location*, change `demo_dir` to `".."` if needed, and source it. Without installing anything: `source("bdscrub_standalone.R")` and then call the functions as below.

3. Rules every function follows

1. **Never guess silently.** A value that cannot be read becomes `NA`, and a warning says how many (switch off with `warn = FALSE`). Ambiguous values (a name used in four districts, two phone numbers in one cell, an income range) are `NA` in the plain result; `details = TRUE` shows what was found and why the value is `NA`.
2. **Report every change.** Place names that were not written in the standard spelling and were changed are listed in a message at the end (`report` argument). Typing-error corrections are marked as guesses.
3. **One value in, one value out.** The result has the same length and order as the input; tables have one row per input value, with the input in the first column.
4. **Bangla works everywhere.** Bangla digits, Bangla place names, Bangla units (lakh, koti, hajar, taka) and the common Unicode variants of the same letters are read by every function.
5. **`details = FALSE` gives a vector, `details = TRUE` a table.** This replaces the many separate functions of version 0.1.

4. Mobile numbers: `bd_phone()`

```
bd_phone(x, format = c("local", "e164", "international"), details = FALSE,
         multiple = c("na", "first"), allow_missing_zero = TRUE, plus881 = FALSE,
         warn = TRUE)
```

Reads Bangladeshi mobile numbers however they were typed and writes them in one format. Spaces, dashes (also Unicode dashes), dots, brackets, quotes and invisible characters are removed; Bangla digits are read; the country code may be written `+880`, `00880` or `880`. A value is a valid number when what remains is a 10-digit national number starting with 13 to 19 (with or without the leading 0). This checks the *format*; it cannot tell whether a number is in service.

Argument	Meaning
x	Character, numeric or factor vector of numbers. Numbers stored as numbers keep all their digits.

Argument	Meaning
format	`"local"` (01712345678, default), `"e164"` (+8801712345678: databases, SMS gateways) or `"international"` (+880 1712-345678: printing).
details	`FALSE`: return the numbers. `TRUE`: return a table with `input`, `phone`, `valid`, `n_numbers`, `all_numbers`, `operator`, `note`.
multiple	A cell can hold several numbers, separated by "/", ",", ";", "&", "|", "or" or line breaks. `"na"` (default) returns `NA` so that no number is chosen silently; `"first"` returns the first. `all_numbers` lists them all either way.
allow_missing_zero	Accept 10-digit numbers without the leading 0 (1712345678)? Spreadsheets drop the 0 when a column is stored as numbers. Default `TRUE`.
plus881	Read "+881" followed by nine digits as "+880" with the 0 dropped? +881 is the satellite-phone code, but "+8817..." is a common typing error for "+88017...". Default `FALSE`; every value read this way is named in `note`.
warn	Warn how many non-missing values gave no number. Default `TRUE`.

```
> x <- c("01712345678", "+880 1712-345678", "8801812345678", "1912345678",
+        "'01712345678", "0171234567", "01712345678 / 01812345678",
+        "01712345678 01912345678", "+881712345678", NA)
> bd_phone(x)
 [1] "01712345678" "01712345678" "01812345678" "01912345678" "01712345678" NA
 [7] NA            NA            NA            NA
Warning message: 2 non-missing values could not be read as mobile numbers and were set to NA.
```

The default gives `NA` for the cell with two numbers. Choosing is an explicit decision:

```
> bd_phone(x, format = "e164", multiple = "first", warn = FALSE)
 [1] "+8801712345678" "+8801712345678" "+8801812345678" "+8801912345678" "+8801712345678"
 [6] NA               "+8801712345678" "+8801712345678" NA               NA
> bd_phone(x[2], format = "international")
[1] "+880 1712-345678"
```

The table explains every result. `operator` is the operator the prefix was *allocated* to; since mobile number portability (2018) a number may now belong to another operator.

```
> bd_phone(x, details = TRUE, warn = FALSE)
                    input       phone valid n_numbers              all_numbers
1               01712345678 01712345678  TRUE         1              01712345678
2           +880 1712-345678 01712345678  TRUE         1              01712345678
3              8801812345678 01812345678  TRUE         1              01812345678
4                 1912345678 01912345678  TRUE         1              01912345678
5               '01712345678 01712345678  TRUE         1              01712345678
6                 0171234567        <NA> FALSE         0                     <NA>
7  01712345678 / 01812345678        <NA>  TRUE         2 01712345678; 01812345678
8    01712345678 01912345678        <NA>  TRUE         2 01712345678; 01912345678
9             +881712345678        <NA> FALSE         0                     <NA>
10                      <NA>        <NA>    NA         0                     <NA>
       operator
1  Grameenphone
2  Grameenphone
3         Robi
4    Banglalink
5  Grameenphone
6          <NA>
7          <NA>
8          <NA>
9          <NA>
10         <NA>
                                                                    note
1                                                                   <NA>
2                                                                   <NA>
3                                                                   <NA>
4                                                                   <NA>
5                                                                   <NA>
6                                                           too few digits
7                          2 numbers in one cell; phone is NA (see `multiple`)
```

```
8                          2 numbers in one cell; phone is NA (see `multiple`)
9  country code +881 (satellite phones); +880 with its 0 missing? See `plus881`
10                                                                        <NA>
```

```
> bd_phone("+8801712345678", plus881 = TRUE)
[1] "01712345678"
> bd_phone("1712345678", allow_missing_zero = FALSE, warn = FALSE)
[1] NA
```

5. Money and numbers

`bd_amount()` — money amounts

```
bd_amount(x, details = FALSE, warn = TRUE)
```

Reads amounts written in Bangladeshi styles and returns them in taka. The reader removes the taka sign, “Tk”, “BDT”, “taka” (also in Bangla), the endings “/-” and “/=” and “only”; removes commas of *correctly* grouped digits (1,25,000 or 125,000); multiplies by thousand (k, hajar), lakh (lac), million, crore (cr, koti) and billion, in English or Bangla; and adds up combined amounts such as “1 crore 25 lakh” or “12 hajar 335”. Amounts with units are computed exactly (1.1 lakh is exactly 110000).

What is refused, and why. Wrongly grouped digits (“Tk 25,00”) could be 2,500 or 25,000, so they are `NA`. “12 hajar 1335” does not add up and is likely a typing error. Ranges (“15-20k”, “2 lakh theke 3 lakh”) and approximate amounts (“about 5 lakh”) are not exact amounts: they are `NA` in the plain result and kept, with their lower and upper end, by `details = TRUE`. Whether to use the midpoint or an interval model is a decision for the analysis, not for data cleaning.

Argument	Meaning
`x`	Character vector of amounts.
`details`	`FALSE`: exact amounts. `TRUE`: table with `input`, `amount` (exact, else `NA`), `low`, `high`, `approximate`, `note`. A range with a unit only at the upper end (“15-20k”) uses that unit for both ends, and `note` says so.
`warn`	Warn how many non-missing values are not exact amounts (with `details = TRUE`: not amounts at all).

```
> inc <- c("Tk 1,25,000", "BDT 3.2 crore", "1.5 lakh", "1 crore 25 lakh", "12 hajar 335",
+          "500/-", "25k", "20000 per month", "15-20k", "about 5 lakh", "Tk 25,00", "N/A")
> bd_amount(inc)
 [1]   125000 32000000   150000 12500000    12335      500    25000    20000       NA
[10]       NA       NA       NA
Warning message: 4 non-missing values could not be read as exact amounts and were set to NA.
```

```
> bd_amount(inc, details = TRUE, warn = FALSE)
             input   amount      low     high approximate
1      Tk 1,25,000   125000   125000   125000       FALSE
2    BDT 3.2 crore 32000000 32000000 32000000       FALSE
3        1.5 lakh   150000   150000   150000       FALSE
4  1 crore 25 lakh 12500000 12500000 12500000       FALSE
5     12 hajar 335    12335    12335    12335       FALSE
6            500/-      500      500      500       FALSE
7              25k    25000    25000    25000       FALSE
8  20000 per month    20000    20000    20000       FALSE
9           15-20k       NA    15000    20000       FALSE
10    about 5 lakh       NA   500000   500000        TRUE
11       Tk 25,00       NA       NA       NA       FALSE
12            N/A       NA       NA       NA       FALSE
                                          note
1                                         <NA>
```

```
2                                          <NA>
3                                          <NA>
4                                          <NA>
5                                          <NA>
6                                          <NA>
7                                          <NA>
8                               "per month" ignored
9  a range; unit of the upper end used for both ends
10                                      approximate
11                        commas in unusual places
12                                       no number
```

```
> bn <- intToUtf8(c(0x09EB, 0x20, 0x09B2, 0x09BE, 0x0996, 0x20, 0x099F, 0x09BE, 0x0995, 0x09BE))
> bn
[1] "৫ লাখ টাকা"
> bd_amount(bn)
[1] 5e+05
```

`bd_number()` — plain numbers

```
bd_number(x, warn = TRUE)
```

For counts and measurements (age, household size, land in decimals). Reads Bangla digits and correct lakh or international grouping; anything with words, units or currency is `NA` on purpose: in an age column, “5 lakh” is an error, not 500000. Use `bd_amount()` for money.

Argument	Meaning
`x`	Character vector.
`warn`	Warn how many non-missing values could not be read.

```
> bd_number(c("1,25,000", "125,000", " 34 ", "-3.5", intToUtf8(c(0x09EA, 0x09E7)),
+             "34 years", "5 lakh", "1,2,5"))
[1] 125000.0 125000.0     34.0     -3.5     41.0       NA       NA       NA
Warning message: 3 non-missing values could not be read as numbers and were set to NA.
```

`bd_taka()` — write amounts

```
bd_taka(x, digits = 0, prefix = "", suffix = "", bangla = FALSE)
```

Argument	Meaning
`x`	Numeric vector.
`digits`	Decimal places. Default 0.
`prefix, suffix`	Text before and after each number, e.g. `prefix = "Tk "`, `suffix = "/-"`.
`bangla`	Write the digits in Bangla. Default `FALSE`.

```
> bd_taka(c(500, 125000, 12500000, -9876543.21), digits = 2)
[1] "500.00"          "1,25,000.00"    "1,25,00,000.00" "-98,76,543.21"
> bd_taka(1250000, prefix = "Tk ", suffix = "/-")
[1] "Tk 12,50,000/-"
> bd_taka(125000, bangla = TRUE)
[1] "১,২৫,০০০"
```

`bd_digits()` — Bangla and English digits

```
bd_digits(x, to = c("en", "bn"))
```

Only needed for text you keep as text (house numbers, IDs); every other function already reads Bangla digits.

Argument	Meaning
x	Character vector.
to	`"en"` (default): Bangla to English digits; `"bn"`: English to Bangla.

```
> bn <- intToUtf8(0x09E6 + c(0, 1, 7, 1, 2, 3, 4, 5, 6, 7, 8))
> bn
[1] "০১৭১২৩৪৫৬৭৮"
> bd_digits(bn)
[1] "01712345678"
> bd_digits("Room 205", to = "bn")
[1] "Room ২০৫"
```

6. Place names: `bd_place()`

```
bd_place(x, level = c("district", "division", "thana"), district = NULL,
         style = c("current", "legacy", "bangla", "map"), fuzzy = TRUE,
         max_distance = 2, details = FALSE, report = c("fixes", "all", "none"),
         warn = TRUE)
```

Looks up place names written in any of their known spellings and returns one standard spelling, so that data from different sources can be joined with each other and with boundary files.

How matching works. (1) Each name is reduced to a *key*: lower case, punctuation as space, words such as District, Zila, Division, City, Thana, P.S., Upazila (also in Bangla) removed, “-gonj” read as “-ganj”; for Bangla, long/short i and u, nukta letters typed as one or two characters, and the two na and two sa letters are made equal. (2) The key is looked up among all known spellings: the official (2018) spelling, the pre-2018 spelling, the boundary-file spelling, the Bangla name, and variants found in survey, administrative and post office data. (3) If nothing is found and `fuzzy = TRUE`, the closest spelling by edit distance is accepted only when it is *one* place, at most `max_distance` edits away, and at most a third of the length of the name (one edit for names of 3–5 letters); spellings of 3 letters or fewer (“Ctg”) are matched exactly only. (4) Changed names are listed in a message.

Argument	Meaning
x	Character vector of names. For districts also codes: BBS (“26” or 26), pcode (“BD3026”), ISO 3166-2 (“BD-13”). Plain numbers are read as BBS codes (BBS 26 is Dhaka; ISO BD-26 is Kishoreganj).
level	`"district"` (default), `"division"` or `"thana"` (thana or upazila names).
district	For thanas only: the district of each value, in any spelling, as one value or one per element (typically the district column of the same data). Decides between places with the same name and makes “Sadar” readable. An upazila not in the given district is not matched.
style	`"current"`: official spelling. `"legacy"`: pre-2018 spelling (for thanas the same as `"map"`). `"bangla"`: Bengali script (districts and divisions only). `"map"`: spelling in the HDX/BBS boundary files, for joining to a map; upazilas created after those files have none (`NA`).
fuzzy	Correct typing errors? Default `TRUE` (new in 1.0.0).

Argument	Meaning
`max_distance`	Largest number of edits for a typing error. Default 2.
`details`	`TRUE`: a table; columns depend on `level` (below). `match` is exact, variant, typing error, code, ambiguous, not found or missing; `matched_as` names the spelling list that matched.
`report`	`"fixes"` (default): list variants and typing errors; `"all"`: also pre-2018 and Bangla spellings; `"none"`.
`warn`	Warn how many non-missing values could not be matched to one place.

Districts

```
> bd_place(c("Chittagong", "jessore district", "Munshigonj", "COX'S BAZAR",
+            "Nababganj", "Brahamanbaria", "Kishorgnj", "Atlantis"))
[1] "Chattogram"       "Jashore"          "Munshiganj"       "Cox's Bazar"
[5] "Chapai Nawabganj" "Brahmanbaria"     "Kishoreganj"      NA
Warning message: 1 non-missing value could not be matched to one district and was set to NA.
3 values were not written as the standard district name and were changed:
  "Kishorgnj" -> "Kishoreganj" (typing error, corrected by fuzzy matching)
  "Nababganj" -> "Chapai Nawabganj" (typing error, corrected by fuzzy matching)
  "Brahamanbaria" -> "Brahmanbaria" (spelling used in boundary (map) files)
Typing-error corrections are guesses: check them by hand.
```

Spelling styles:

```
> bd_place(c("Chattogram", "Bogura", "Brahmanbaria"), style = "legacy")
[1] "Chittagong"   "Bogra"        "Brahmanbaria"
> bd_place(c("Chattogram", "Bogura", "Brahmanbaria"), style = "map")
[1] "Chittagong"    "Bogra"         "Brahamanbaria"
> bd_place(c("Jessore", "Bogra"), style = "bangla")
[1] "যশোর" "বগুড়া"
```

`report`, `fuzzy` and `max_distance`:

```
> bd_place(c("Chittagong", "Comilla"), report = "all")
[1] "Chattogram" "Cumilla"
2 values were not written as the standard district name and were changed:
  "Chittagong" -> "Chattogram" (spelling used before 2018)
  "Comilla" -> "Cumilla" (spelling used before 2018)
> bd_place(c("Kishorgnj", "Sylht"), fuzzy = FALSE, warn = FALSE)
[1] NA NA
> bd_place("Sylht", max_distance = 0, warn = FALSE)
[1] NA
```

Details and codes:

```
> bd_place(c("Comilla", "26", "BD-13", "BD4041", "Ctg"), details = TRUE)
    input   district   division bbs_code  pcode iso_code   match   matched_as candidates
1 Comilla    Cumilla Chattogram       19 BD2019    BD-08 variant old spelling       <NA>
2      26      Dhaka      Dhaka       26 BD3026    BD-13    code         <NA>       <NA>
3   BD-13      Dhaka      Dhaka       26 BD3026    BD-13    code         <NA>       <NA>
4  BD4041    Jashore     Khulna       41 BD4041    BD-22    code         <NA>       <NA>
5     Ctg Chattogram Chattogram       15 BD2015    BD-10 variant      variant       <NA>
1 value was not written as the standard district name and was changed:
  "Ctg" -> "Chattogram" (not the standard spelling)
```

Divisions

```
> bd_place(c("Barisal Division", "chittagong div", "Rongpur", "Dacca"), level = "division")
[1] "Barishal"   "Chattogram" "Rangpur"    "Dhaka"
2 values were not written as the standard division name and were changed:
```

```
  "Dacca" -> "Dhaka" (not the standard spelling)
  "Rongpur" -> "Rangpur" (not the standard spelling)
```

Thanas

Below the district, bdscrub works with **thanas** (police stations) only. A rural thana has the same name and area as its upazila, so a column of upazila names is cleaned with `level = "thana"` too, and "Upazila", "Thana" and "P.S." after a name are ignored. `bd_thanas` has three types: **rural** (503), **city** (61 thanas of the city corporations, such as Mirpur and Double Mooring) and **within** (6 thanas inside the area of a larger rural thana: Ashulia in Savar, Fatullah and Siddhirganj in Narayanganj Sadar, South Keraniganj in Keraniganj, Bhujpur in Fatikchhari, Tongi in Gazipur Sadar).

Many thana names occur in several districts (Kaliganj is in four; there is a Kotwali in several cities; Mirpur, Kishoreganj, Faridpur and Sherpur are also thanas outside the districts of the same name). Without `district` such names are "ambiguous" and `NA`; "Sadar" alone is understood only with a district.

```
> bd_place(c("Kaliganj", "Kaliganj", "Sadar", "Iswardi", "Savr", "Teknaf", "Savar Upazila"),
+          level = "thana", district = c("Gazipur", NA, "Feni", "Pabna", "Dhaka", NA, NA))
[1] "Kaliganj"   NA           "Feni Sadar" "Ishwardi"   "Savar"      "Teknaf"
[7] "Savar"
Warning message: 1 non-missing value could not be matched to one thana and was set to NA.
3 values were not written as the standard thana name and were changed:
  "Savr" -> "Savar" (typing error, corrected by fuzzy matching)
  "Iswardi" -> "Ishwardi" (spelling used by the post office)
  "Sadar" -> "Feni Sadar" ("Sadar" read with the district given)
Typing-error corrections are guesses: check them by hand.
```

```
> bd_place(c("Kaliganj", "Sadar"), level = "thana", details = TRUE, warn = FALSE)
     input thana district division type map_name pcode     match matched_as
1 Kaliganj  <NA>     <NA>     <NA> <NA>     <NA>  <NA> ambiguous       <NA>
2    Sadar  <NA>     <NA>     <NA> <NA>     <NA>  <NA> ambiguous       <NA>
                                                                 candidates
1 Kaliganj (Gazipur); Kaliganj (Jhenaidah); Kaliganj (Satkhira); Kaliganj (Lalmonirhat)
2                                                    "Sadar": give the district
  note
1 <NA>
2 <NA>
```

For joining to a map, `style = "map"` gives the boundary-file name. A thana created after the boundary files has none; a "within" thana gets the area it lies in, and a message says so:

```
> bd_place(c("Brahmanbaria Sadar", "Ashulia", "Eidgaon"), level = "thana", style = "map", warn = FALSE)
[1] "Brahmanbaria Sadar" "Savar"              NA
1 thana has no area of its own in the boundary files and was given the area it lies in:
  "Ashulia" -> "Savar"
```

Details:

```
> bd_place(c("Ashulia", "Fatulla", "Savar", "Mirpur", "Double Mooring", "Kotwali"),
+          level = "thana", district = c("Dhaka", "Narayanganj", "Dhaka", "Dhaka", "Chittagong", NA),
+          details = TRUE)
          input          thana    district   division   type          map_name  pcode
1       Ashulia       Ashulia       Dhaka      Dhaka within             Savar   <NA>
2       Fatulla      Fatullah Narayanganj      Dhaka within Narayanganj Sadar   <NA>
3         Savar         Savar       Dhaka      Dhaka  rural             Savar 302672
4        Mirpur        Mirpur       Dhaka      Dhaka   city            Mirpur 302648
5 Double Mooring Double Mooring  Chattogram Chattogram   city    Double Mooring 201528
6       Kotwali          <NA>        <NA>       <NA>   <NA>              <NA>   <NA>
      match matched_as                                                  candidates
1     exact   official                                                       <NA>
2   variant    variant                                                       <NA>
3     exact   official                                                       <NA>
4     exact   official                                                       <NA>
5     exact   official                                                       <NA>
6 ambiguous       <NA> Kotwali (Chattogram); Kotwali (Dhaka); Jashore Sadar (Jashore)
                          note
1            part of the Savar area
```

```
2 part of the Narayanganj Sadar area
3                              <NA>
4        Dhaka city corporation area
5   Chattogram city corporation area
6                              <NA>
Warning message: 1 non-missing value could not be matched to one thana and was set to NA.
1 value was not written as the standard thana name and was changed:
  "Fatulla" -> "Fatullah" (not the standard spelling)
```

7. Addresses: `bd_address()`

```
bd_address(x, district = NULL)
```

Reads the location from free-text addresses. Every value comes from the package lists; nothing is invented.

1. **Post code**: a 4-digit number that is a known post code (one after “code” or at the end is preferred).
2. **Place names**: every sequence of 1–4 words is looked up as a district, division and thana name (upazila names included) in all known spellings. A name followed by Road, College, Medical, Bus stand … is part of a street or institution name and is skipped.
3. **Post office**: without a post code, a post office name after “P.O.” gives one.
4. **District**: with several district names, the one that has a thana or the post code in it, or is named last, wins; a district name that is also a thana of another district named (“Faridpur, Pabna”) is read as the thana. Without a district name: from the thana (if only one district has it) or the post code.
5. **Thana** in that district: the last one named (but a thana inside the area of another named thana wins: “Ashulia, Savar” is Ashulia), else “Sadar”, else the post code’s thana.
6. **Checks**: a division or post code that does not fit the district gives status “conflict”.

Fuzzy matching is not used inside addresses: in a whole address many words would match some place by chance.

Argument	Meaning
`x`	Character vector of addresses.
`district`	Optional district of each address (a separate column, any spelling; one value or one per address). It is used as the district, settles names used in several districts, and an address naming a different district gets status “conflict”.

Result columns: `input`, `postcode`, `post_office`, `thana` (rural, city or “within” thana; see `bd_thanas`), `district`, `division`, `status` (complete, district only, division only, ambiguous, conflict, not found, missing) and `note` (how the result was reached when not direct, and what contradicts what).

```
> a <- c("House 12, Road 5, Mirpur 10, Dhaka-1216",
+        "Vill: Char Kalia, P.O: Kalia, Narail - 7520",
+        "Ashulia, Savar, Dhaka", "Kaliganj", "Kaliganj, Gazipur",
+        "Faridpur, Pabna", "Sarail, B. Baria", "Dhaka Road, Mymensingh",
+        "Mirpur, Dhaka-7030", "Rajshahi Division", "Nowhere street")
> bd_address(a)
                                    input postcode post_office    thana     district
1      House 12, Road 5, Mirpur 10, Dhaka-1216     1216  Mirpur TSO   Mirpur        Dhaka
2  Vill: Char Kalia, P.O: Kalia, Narail - 7520     7520   Kalia UPO    Kalia       Narail
3                       Ashulia, Savar, Dhaka     <NA>        <NA>  Ashulia        Dhaka
4                                   Kaliganj     <NA>        <NA>     <NA>         <NA>
5                           Kaliganj, Gazipur     <NA>        <NA> Kaliganj      Gazipur
6                             Faridpur, Pabna     <NA>        <NA> Faridpur        Pabna
7                              Sarail, B. Baria     <NA>        <NA>   Sarail Brahmanbaria
8                       Dhaka Road, Mymensingh     <NA>        <NA>     <NA>   Mymensingh
9                           Mirpur, Dhaka-7030     7030  Mirpur UPO   Mirpur        Dhaka
10                           Rajshahi Division     <NA>        <NA>     <NA>         <NA>
11                              Nowhere street     <NA>        <NA>     <NA>         <NA>
```

```
     division        status
1       Dhaka      complete
2      Khulna      complete
3       Dhaka      complete
4         <NA>     ambiguous
5       Dhaka      complete
6    Rajshahi      complete
7  Chattogram      complete
8  Mymensingh district only
9       Dhaka      conflict
10   Rajshahi division only
11        <NA>     not found
                                                                                                                                                note
1                                                                                                                                               <NA>
2                                                                                                                                               <NA>
3                                                                                                                                               <NA>
4  thana name used in several districts: Kaliganj (Gazipur); Kaliganj (Jhenaidah); Kaliganj (Satkhira); Kaliganj (Lalmonirhat)
5                                                                                                                                               <NA>
6                                                                                                  several district names (Faridpur, Pabna); used Pabna
7                                                                                                                                               <NA>
8                                                                                                                                               <NA>
9                                                                                                  post code 7030 (Mirpur UPO) is in Kushtia, not Dhaka
10                                                                                                                                              <NA>
11                                                                                                                                              <NA>
```

With a district column:

```
> bd_address(c("Kaliganj", "Bazar road, Kaliganj", "Teknaf"),
+            district = c("Jhenaidah", "Satkhira", "Chittagong"))
                input postcode post_office    thana   district   division        status
1             Kaliganj     <NA>        <NA> Kaliganj  Jhenaidah     Khulna      complete
2 Bazar road, Kaliganj     <NA>        <NA> Kaliganj   Satkhira     Khulna      complete
3               Teknaf     <NA>        <NA>     <NA> Chattogram Chattogram district only
                                                                                    note
1                                                                                    <NA>
2                                                                                    <NA>
3 district from the thana name; names a thana of another district: Teknaf (Cox's Bazar)
```

8. Missing values and spaces

`bd_missing()`

```
bd_missing(x, extra = NULL, markers = NULL, blank = TRUE)
```

Turns whole-value missing markers into `NA` (case and spaces do not matter; the Bangla markers for “not applicable” and “no information” are included). “none”, “unknown” and “don’t know” are *not* treated as missing by default, because in a survey they are often real answers.

Argument	Meaning
`x`	Vector (character, factor, numeric). Factors stay factors; marker levels are dropped.
`extra`	More markers, e.g. `c("none", "unknown")`.
`markers`	Markers to use. `NULL` (default) = the built-in list `bd_na_words`; give a vector to replace it.
`blank`	Empty and space-only values become `NA`. Default `TRUE`.

```
> x <- c("23", "N/A", " - ", "#N/A", "none", "", "not applicable", "N/A at time of survey", NA)
> bd_missing(x)
[1] "23"                    NA                      NA
[4] NA                      "none"                  NA
[7] NA                      "N/A at time of survey" NA
> bd_missing(x, extra = c("none"))
[1] "23"                    NA                      NA
[4] NA                      NA                      NA
[7] NA                      "N/A at time of survey" NA
```

```
> bd_missing(x, markers = character(0), blank = FALSE)
[1] "23"                    "N/A"                   " - "
[4] "#N/A"                  "none"                  ""
[7] "not applicable"        "N/A at time of survey" NA
```

`bd_squish()`

```
bd_squish(x, strip_quotes = FALSE)
```

Collapses spaces, tabs and line breaks (also non-breaking, thin and zero-width spaces and the byte-order mark at the start of CSV files).

Argument	Meaning
`x`	Character vector.
`strip_quotes`	Also remove quotes around the whole value, such as Excel's leading apostrophe. Default `FALSE`.

```
> bd_squish(c("  Dhaka   District ", paste0("Tk", intToUtf8(0xA0), "500"),
+             paste0(intToUtf8(0xFEFF), "01712345678"), "'01712345678"))
[1] "Dhaka District" "Tk 500"         "01712345678"    "'01712345678"
> bd_squish("'01712345678", strip_quotes = TRUE)
[1] "01712345678"
```

9. Whole data frames

`bd_check()`

```
bd_check(data, spec, details = FALSE, district = NULL, id = NULL, markers = NULL,
         fuzzy = TRUE)
```

Reports, for each column in `spec`, how many values are missing, readable and not readable, without changing the data. Missing markers count as missing, not invalid. A phone cell with two numbers, an income range and an ambiguous place count as invalid, because each needs a decision. An address is valid when its district is found and nothing contradicts it.

Argument	Meaning
`data`	A data frame.
`spec`	Named character vector (or list): column = type. Types: `"phone"`, `"amount"`, `"number"`, `"district"`, `"division"`, `"thana"`, `"address"`.
`details`	`FALSE`: one row per column (`n`, `n_missing`, `n_valid`, `n_invalid`, `pct_valid`, `invalid_examples`). `TRUE`: one row per failing value with `row`, `column`, `type`, `value`, `reason`.
`district`	**Name of the column** that holds each row's district; used for thana and address columns.
`id`	Name of an identifier column to include in the details.
`markers`	Missing markers, as in `bd_missing()`; `character(0)` = only `NA` and blanks.
`fuzzy`	Correct typing errors in place names. Default `TRUE`.

```
> df <- data.frame(
+   id      = c("a", "b", "c", "d", "e"),
+   mobile  = c("01712345678", "+880 1812-345678", "12345", "N/A", "01712345678 / 01812345678"),
+   zila    = c("Chittagong", "Jessore", "Dhaka", "Atlantis", "Nababganj"),
+   thana   = c("Double Mooring", "Kotwali", "Ashulia", "Savar", "Shibganj"),
+   income  = c("Tk 25,000", "1.5 lakh", "15-20k", "30000", "-"),
+   address = c("Agrabad, Chattogram 4100", "Kotwali, Jessore", "Ashulia, Savar", "Nowhere", NA),
+   stringsAsFactors = FALSE
+ )
> spec <- c(mobile = "phone", zila = "district", thana = "thana", income = "amount",
+           address = "address")
> bd_check(df, spec, district = "zila")
   column     type n n_missing n_valid n_invalid pct_valid                invalid_examples
1  mobile    phone 5         1       2         2        50 12345; 01712345678 / 01812345678
2    zila district 5         0       4         1        80                         Atlantis
3   thana    thana 5         0       5         0       100
4  income   amount 5         1       3         1        75                           15-20k
5 address  address 5         1       3         1        75                          Nowhere
Warning message: 1 value of `district` could not be matched and was ignored: Atlantis
```

```
> bd_check(df, spec, details = TRUE, district = "zila", id = "id")
  id row  column     type                    value
1  c   3  mobile    phone                    12345
2  e   5  mobile    phone 01712345678 / 01812345678
3  d   4    zila district                  Atlantis
4  c   3  income   amount                    15-20k
5  d   4 address  address                   Nowhere
                                             reason
1                                  too few digits
2 2 numbers in one cell; phone is NA (see `multiple`)
3                               not a district name
4   a range; unit of the upper end used for both ends
5                                       not found
Warning message: 1 value of `district` could not be matched and was ignored: Atlantis
```

`bd_clean()`

```
bd_clean(data, spec, district = NULL, suffix = "_clean",
         phone_format = c("local", "e164", "international"),
         style = c("current", "legacy", "bangla", "map"), fuzzy = TRUE,
         report = c("fixes", "all", "none"), markers = NULL)
```

Cleans each column in `spec` and adds the results as new columns, keeping the originals: phone, amount, number, district, division, thana → `<column>_clean`; address → `<column>_postcode`, `_thana`, `_district`, `_division`, `_status`. Changed place names are reported per column.

Argument	Meaning
`data, spec, district, fuzzy, markers`	As in `bd_check()`.
`suffix`	Added to the column name. `""` replaces the original column.
`phone_format`	As `format` in `bd_phone()`.
`style`	As in `bd_place()`.
`report`	As in `bd_place()`.

```
> out <- bd_clean(df, spec, district = "zila")
Warning message: 1 value of `district` could not be matched and was ignored: Atlantis
zila: 1 value was not written as the standard district name and was changed:
  "Nababganj" -> "Chapai Nawabganj" (typing error, corrected by fuzzy matching)
Typing-error corrections are guesses: check them by hand.
thana: 1 value was not written as the standard thana name and was changed:
  "Kotwali" -> "Jashore Sadar" (spelling used in boundary (map) files)
> names(out)
 [1] "id"               "mobile"           "zila"             "thana"
 [5] "income"           "address"          "mobile_clean"     "zila_clean"
 [9] "thana_clean"      "income_clean"     "address_postcode" "address_thana"
[13] "address_district" "address_division" "address_status"
> out[, c("mobile_clean", "zila_clean", "thana_clean", "income_clean")]
  mobile_clean       zila_clean    thana_clean income_clean
1  01712345678       Chattogram Double Mooring        25000
2  01812345678          Jashore  Jashore Sadar       150000
```

```
3         <NA>            Dhaka        Ashulia           NA
4         <NA>             <NA>          Savar        30000
5         <NA> Chapai Nawabganj       Shibganj           NA
> out[, c("address_postcode", "address_thana", "address_district", "address_status")]
  address_postcode address_thana address_district address_status
1             4100          <NA>       Chattogram  district only
2             <NA> Jashore Sadar          Jashore       complete
3             <NA>       Ashulia            Dhaka       complete
4             <NA>          <NA>             <NA>      not found
5             <NA>          <NA>             <NA>        missing
```

10. Data sets

Data set	Rows	Content and source
`bd_districts`	64	current, pre-2018, boundary-file and Bangla names; division; BBS, pcode and ISO 3166-2 codes (ISO 3166-2:BD, BBS, Prothom Alo 2018).
`bd_divisions`	8	names and codes.
`bd_thanas`	570	503 rural thanas (same name and area as the upazila; HDX/BBS boundaries via the CRAN package ‘bangladesh’, reconciled with Wikipedia’s list, Oct 2026), 61 city and 6 “within” thanas; type, map name, pcode.
`bd_postcodes`	1,329	post offices and post codes from Bangladesh Post district pages (Oct 2026), linked to thanas (1,260 linked); postal and administrative district.
`bd_phone_prefixes`	7	prefixes 013–019 and allocated operator (BTRC, 2018).
`bd_na_words`	24	missing-value markers and their meaning.

```
> head(bd_districts, 3)
  district district_legacy district_map district_bn division bbs_code  pcode iso_code
1  Barguna         Barguna      Barguna       বরগুনা Barishal       04 BD1004    BD-02
2 Barishal         Barisal      Barisal      বরিশাল Barishal       06 BD1006    BD-06
3    Bhola           Bhola        Bhola        ভোলা Barishal       09 BD1009    BD-07
> bd_thanas[c(1, 504:507, 511), ]
              thana    district   division   type          map_name  pcode
1            Amtali     Barguna   Barishal  rural            Amtali 100409
504         Fatullah Narayanganj      Dhaka within Narayanganj Sadar
505      Siddhirganj Narayanganj      Dhaka within Narayanganj Sadar
506          Ashulia       Dhaka      Dhaka within             Savar
507 South Keraniganj       Dhaka      Dhaka within        Keraniganj
511  Bayejid Bostami  Chattogram Chattogram   city   Bayejid Bostami 201506
                                note        source
1                                    map+wikipedia
504 part of the Narayanganj Sadar area        manual
505 part of the Narayanganj Sadar area        manual
506            part of the Savar area        manual
507        part of the Keraniganj area        manual
511   Chattogram city corporation area           map
> head(bd_postcodes, 3)
  postcode    post_office postal_thana postal_district     thana district division note
1     1000      Dhaka GPO    Motijheel           Dhaka Motijheel    Dhaka    Dhaka
2     1100 Dhaka Sadar HO  Dhaka Sadar           Dhaka              Dhaka    Dhaka
3     1203       Wari TSO   Dhaka Sadr           Dhaka              Dhaka    Dhaka
```

11. The messy demonstration data

`data/bdscrub_messy_survey.csv` has 3,030 rows (3,000 records and 30 exact duplicates, as when a form is submitted twice) and the columns `id`, `name`, `mobile`, `age`, `monthly_income`, `division`, `district`, `thana` (written as thana or upazila), `address`. Each row starts from a true record drawn from the package tables (a district, thana and post code that belong together; 80% rural thanas, 5% “within” thanas, 15% city thanas), and a made-up person (random first name and surname; random phone numbers). The true values are in `data/bdscrub_answer_key.csv`, with a column saying how each cell was written (e.g. `district_written_as` = exact, old spelling, suffix, Bangla, variant, map

spelling, typing error, missing). The messiness: numbers with +880, dashes, dropped 0, Bangla digits, Excel apostrophes, two numbers per cell, +881 errors, too short, wrong prefix, scientific notation; incomes with Tk, BDT, /-, k, lakh, hajar, Bangla, ranges, “about”, wrong commas, “unknown”; place names in every spelling listed above, with one-letter typing errors in names of 7+ letters; addresses in seven templates with and without post office and post code.

Results of `demo/demo_bdscrub.R` (run on the package code; the same with the one-file version):

Field	correct	flagged (NA)	wrong
mobile (`multiple = "first"`)	100.0%	0.0%	0.0%
age	100.0%	0.0%	0.0%
income, exact / lower / upper end	100.0%	0.0%	0.0%
division, district	100.0%	0.0%	0.0%
thana, district column given	99.5%	0.5%	0.0%
thana, no district	94.7%	5.3%	0.0%
address → district (2,932 addresses with a place)	99.9%	0.1%	0.0%
address → thana (where one was found)	100.0%	0.0%	0.0%

> **How to read these numbers.** “Correct” includes cells that are correctly `NA` (a missing marker, an invalid number). The messy data were generated from the *same* spelling lists the package uses, plus random one-letter typing errors, so these scores show that the functions are consistent and that the safeguards work (no wrong values, ambiguous names refused); they are **not** an estimate of accuracy on real survey data, where unseen spellings, multi-letter errors and village names that equal thana names elsewhere will occur. Validation on 200–500 real, hand-coded addresses is still needed before a CRAN release.

12. What changed from 0.1, limitations, and what is not verified

Old names → new calls

0.1 / development function	1.0.0
`bd_phone_standardise(x, format)`	`bd_phone(x, format)`
`bd_phone_valid(x)`	`bd_phone(x, details = TRUE)$valid`
`bd_phone_prefix_operator(x)`	`bd_phone(x, details = TRUE)$operator`
`bd_phone_extract(x)`	`bd_phone(x, details = TRUE, multiple = "first")`
`bd_parse_amount(x)`, `bd_parse_amount_range(x)`	`bd_amount(x)`, `bd_amount(x, details = TRUE)`
`bd_parse_number()`, `bd_format_amount()`	`bd_number()`, `bd_taka()`
`bn_to_en_digits(x)`, `en_to_bn_digits(x)`	`bd_digits(x)`, `bd_digits(x, to = "bn")`

0.1 / development function	1.0.0
`bd_district_standardise(x)`, `bd_division_standardise(x)`	`bd_place(x)`, `bd_place(x, level = "division")`
`bd_district_info(x)`	`bd_place(x, details = TRUE)`
`bd_upazila_match()`, `bd_upazila_standardise()`	`bd_place(x, level = "thana", district = ...)`
`bd_extract_location(x)`	`bd_address(x)`
`bd_na_if()`, `bd_na_markers()`	`bd_missing()`, `bd_na_words`
`bd_failures(df, spec)`	`bd_check(df, spec, details = TRUE)`
`bd_has_bangla()`	removed

Behaviour changes: `fuzzy = TRUE` is the default; a phone cell with two numbers is `NA` unless `multiple = "first"`; a number inside other text ("Mobile: 017…") is now found.

Changes of 8 November 2026. Below the district only **thanas** are kept: the `"upazila"` level of `bd_place()`, the `upazila` column of `bd_address()`, the `"upazila"` type of `bd_check()`/`bd_clean()` and the data set `bd_upazilas` were removed; upazila names are read as thana names. Review fixes, each with a regression test:

- `bd_phone()`: numbers separated only by spaces ("01712345678 01812345678") or words are all found; numbers stored by a spreadsheet with ".0" ("8801712345678.0") are read.
- `bd_amount()`: "per month", "monthly", "/month", "yearly" (also Bangla) after an amount are ignored and named in `note`; a dash after the currency ("BDT-500", "Tk.-500") is a separator, not a minus sign (it was read as −500).
- `bd_place()`: "City" and "City Corporation" after a name are ignored ("Dhaka City").
- `bd_address()`: "N/A" and other markers give status "missing" (was "not found"); when an address names a thana and the larger thana it lies in, the smaller one is used in either order ("Savar, Ashulia" gave Savar).
- `bd_clean()` with `style = "map"` now reports thanas given the map area they lie in.
- Data: Tongi added as a thana (inside Gazipur Sadar); two post office names with typing errors on the Bangladesh Post pages corrected ("Dhaka Cen,ft. TSO", "Brah,anbaria HO").

Known gaps

- Post codes of Mymensingh division are missing (the Bangladesh Post page was not found); 69 post offices are not linked to a thana.
- Bangla thana names are not in the data yet (`style = "bangla"` gives `NA` for them).
- Newer metropolitan thanas (for example of Dhaka after the boundary files) are missing; the "within" list has 5 thanas and is not complete.
- The 20 rural thanas taken from Wikipedia only (created after the boundary files) are not yet checked against the government gazette.
- The reuse terms of the Bangladesh Post list and of the boundary names have not been confirmed; this matters before CRAN.

What was verified, and how

- 285 test expectations (ported 0.1 tests, 47 hand-made address cases, thana, phone, amount, data-frame tests and the regression tests of the 8 November review) pass, and all help-page examples run, in WebAssembly R (webR) in the build environment.

- The demonstration script runs end-to-end on the package code and on `bdscrub_standalone.R`.
- **`R CMD check --as-cran --no-manual` on R 4.6.1, Windows 11 (8 November 2026, after the thana-only change and the review fixes): Status OK**, with 0 errors, 0 warnings and 0 notes. The documentation was generated by roxygen2, and the tests, examples and vignette were run with tibble installed (`check_log.txt` in the folder). The online CRAN incoming check was switched off because the computer could not reach CRAN, and the PDF manual was not built. Both must be run before submission.
- The demo also ran with the installed package in RStudio, with the same scores.